

Scenario Based Quiz

Deep Learning, CNNs & Computer Vision

Scenario 1: Defect Detection in a Factory

A smartphone manufacturing company wants to **automatically detect scratches on phone screens** using images captured from a camera on the assembly line.

Q1.

The images captured are in **RGB format** and need to be processed before training the model. Which OpenCV function would you use to **read the image into Python**?

- A. `cv2.open()`
 - B. `cv2.read()`
 - C. `cv2.imread()`
 - D. `cv2.load()`
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Q2.

Before feeding images into the CNN, the engineer converts them to **grayscale** to reduce computational complexity.

Which OpenCV function is used for this transformation?

- A. `cv2.resize()`
 - B. `cv2.cvtColor()`
 - C. `cv2.threshold()`
 - D. `cv2.blur()`
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Q3.

To highlight scratches on the screen edges, the engineer decides to use **edge detection**.

Which algorithm is most commonly used?

- A. Gaussian Blur
 - B. Canny Edge Detection
 - C. Median Filter
 - D. Morphological Closing
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Scenario 2: CNN Feature Learning

A data scientist builds a **CNN model to classify images of cats and dogs**.

Q4.

The **convolution layer** in a CNN mainly performs which function?

- A. Reduces the image size
 - B. Extracts spatial features such as edges and textures
 - C. Converts images to grayscale
 - D. Stores the training data
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Q5.

After convolution, the model applies **Max Pooling**.

What is the main purpose of pooling?

- A. Increase the image size
 - B. Reduce dimensionality and retain important features
 - C. Increase the number of filters
 - D. Normalize the image pixels
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Q6.

Which of the following represents a **typical CNN architecture order**?

- A. Pooling → Fully Connected → Convolution
- B. Convolution → Pooling → Fully Connected

- C. Fully Connected → Convolution → Pooling
 - D. Pooling → Convolution → Softmax
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Scenario 3: Training an Image Classifier

A team is training a CNN model to classify **plant diseases from leaf images**.

Q7.

Which of the following is **commonly used as the loss function** for multi-class image classification?

- A. Mean Squared Error
 - B. Binary Cross Entropy
 - C. Categorical Cross Entropy
 - D. Hinge Loss
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Q8.

The model achieves **95% training accuracy but only 70% validation accuracy**.

This is an example of:

- A. Underfitting
 - B. Overfitting
 - C. Perfect model
 - D. Data normalization
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Q9.

Which technique helps **improve model accuracy and prevent overfitting**?

- A. Increasing batch size only
 - B. Data augmentation
 - C. Removing convolution layers
 - D. Reducing dataset size
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Scenario 4: Transfer Learning

A startup wants to build an **animal recognition app** but has only **200 labeled images**.

Q10.

Which technique is most appropriate in this case?

- A. Train CNN from scratch
 - B. Transfer learning using pretrained models**
 - C. Use linear regression
 - D. Reduce image resolution
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Q11.

Which pretrained CNN model is commonly used for **transfer learning**?

- A. K-Means
 - B. ResNet**
 - C. PCA
 - D. Apriori
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Scenario 5: Real World Computer Vision Application

A city government installs cameras to **automatically detect traffic violations**.

Q12.

Which computer vision task is most relevant?

- A. Image segmentation
- B. Image classification / object detection**
- C. Speech recognition
- D. Text summarization

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